Citations and Evaluation Guide

Why page metadata matters

Every chunk stored in the vector database should carry the name of its source file and the page number it came from. This metadata is what allows the assistant to print a citation next to each claim and allows a reader to open the original file at the right page to verify the claim. Without page metadata a citation can only point to a whole document, which makes verification slow and discourages readers from checking the answer at all.

Citation format and validation

Citations follow a stable pattern made of the file name, a comma, the abbreviation p. and the page number, all wrapped in square brackets. A stable pattern can be parsed with a regular expression after generation. Language models occasionally invent citations or copy a reference that never appeared in the evidence. For that reason the application should validate the answer: every citation found in the generated text must match one of the file and page pairs that were actually retrieved. Citations that fail this check are reported to the user as unverified.

Evaluating retrieval and generation separately

Retrieval and generation fail in different ways, so they should be evaluated separately. Retrieval quality is measured with recall at k: for a set of test questions with known answer pages, count how often the correct page appears among the top k results. Generation quality is measured by faithfulness: whether every statement in the answer is supported by the retrieved passages. A third check covers refusals: questions whose answer is not in the corpus must produce the refusal sentence. A small set of ten to twenty hand written questions is enough to compare chunk sizes, thresholds and models.